

EXACT COUNTEREXAMPLES TO AN EHRHART ROOT-DISK CONJECTURE FOR GENERALIZED SNAKE POSETS

ZIJIAN ZENG

ABSTRACT. Lee, Vindas-Meléndez, and Wang conjectured a disk bound for the roots of Ehrhart polynomials of generalized-snake order polytopes. The printed statement has a sign error and an off-by-one length parameter; we use the unique repair compatible with their definition and their proved functional equation. We give exact computer-assisted counterexamples for the regular snake words of lengths 9 and 10. For length 10, the requested next finite layer, the word $\varepsilon LRLRLRLRLR$ has a root ρ satisfying $|\rho + 7| > 6$. A rational Rouché disk proves the strict inequality without floating-point roots. A separate verifier reconstructs the Ehrhart polynomial from order-ideal multichains and certifies the same failure through an exact Cayley transform and Routh table. The length-9 example also shows that the authors’ reported computation through length 9 is incompatible with the corrected statement.

1. INTRODUCTION

For a generalized snake word $\mathbf{w}$, Lee, Vindas-Meléndez, and Wang studied the Ehrhart polynomial $L(\mathbf{w}; t)$ of the order polytope $\mathcal{O}(P(\mathbf{w}))$ [2]. Their Conjecture 5.1 has two parts. The first, real-rootedness of the h^* -polynomial, has since been proved by Braun and Jal [1]. We concern ourselves only with the second, an Ehrhart-root disk bound.

There is a necessary normalization issue. Definition 2.1 of [2] calls $\varepsilon w_1 \cdots w_m$ a word of length m , whereas Conjecture 5.1 says “length $n + 1$.” The displayed disk has positive center even though the same line states a negative symmetry axis. Theorem 2.9 of that paper proves

$$(1) \quad \prod_{j=1}^{m+3} (t + j) \mid L(\mathbf{w}; t), \quad L(\mathbf{w}; t) = L(\mathbf{w}; -m - 4 - t).$$

Accordingly, the only repair compatible simultaneously with the definition, the forced roots, and the functional equation is

$$(2) \quad L(\mathbf{w}; \rho) = 0 \implies \left| \rho + \frac{m+4}{2} \right| \leq \frac{m+2}{2} \quad (\mathbf{w} \text{ of actual length } m).$$

This is also the interpretation specified in the research question that led to the present work. We do not assign mathematical content to the inconsistent literal display.

The authors report having checked their conjecture for words of length at most 9 [2, after Conjecture 5.1], but no code or exact certificate accompanies that sentence. Exact reconstruction gives the following result.

Theorem 1.1. *Under the repaired statement (2):*

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- (1) the regular snake word $\mathbf{w}_9 = \varepsilon LRLRLRLRL$ of length 9 has an Ehrhart root ρ_9 with $|\rho_9 + 13/2| > 11/2$;
- (2) the regular snake word $\mathbf{w}_{10} = \varepsilon LRLRLRLRLR$ of length 10 has an Ehrhart root ρ_{10} with $|\rho_{10} + 7| > 6$.

Consequently the corrected all-length conjecture and, in particular, the universal length-10 statement are false.

The proof uses only integer and rational arithmetic after a witness word is specified. Decimal approximations are included only for orientation. The complete certificates, reconstructing verifiers, and rejection tests are described in Section 5.

2. DEFINITIONS AND RADIAL REDUCTION

Write a generalized snake word as

$$\mathbf{w} = \varepsilon w_1 \cdots w_m, \quad w_i \in \{L, R\}.$$

The poset $P(\varepsilon)$ has elements $\{0, 1, 2, 3\}$ and covers $3 \prec 1, 3 \prec 2, 1 \prec 0, 2 \prec 0$. At step $k \geq 1$, adjoin $2k+2, 2k+3$ and the covers

$$2k+3 \prec 2k+1, \quad 2k+3 \prec 2k+2,$$

together with $2k+2 \prec 2k-1$ if $k=1, w_k = L$, or if $k \geq 2, w_{k-1} \neq w_k$; otherwise adjoin $2k+2 \prec 2k$. Thus $P(\mathbf{w})$ has $d = 2m+4$ elements. Its order polytope is

$$\mathcal{O}(P) = \{x \in [0, 1]^P : x_i \leq x_j \text{ whenever } i <_P j\}.$$

We normalize the Ehrhart numerator by

$$(3) \quad L(\mathbf{w}; t) = \sum_{i=0}^d h_i^* \binom{t+d-i}{d}, \quad h^*(\mathbf{w}; z) = \sum_i h_i^* z^i.$$

The source factorization and symmetry reduce the non-forced roots to a much smaller radial polynomial.

Lemma 2.1. *Set $x = t + (m+4)/2$ and $A_m(t) = \prod_{j=1}^{m+3} (t+j)$. There is a real polynomial R_m of degree $\lfloor (m+1)/2 \rfloor$ such that*

$$L\left(x - \frac{m+4}{2}\right) = A_m\left(x - \frac{m+4}{2}\right) x^\epsilon R_m(x^2), \quad \epsilon \equiv m+1 \pmod{2}.$$

Up to the fixed roots, the disk assertion is equivalent to requiring every zero y of R_m to satisfy

$$|y| \leq \left(\frac{m+2}{2}\right)^2.$$

Proof. Equation (1) makes the shifted Ehrhart polynomial even. The shifted fixed factor has parity $(-1)^{m+3}$. Its quotient therefore has degree $m+1$ and parity $(-1)^{m+1}$, proving the displayed form. The fixed roots have $|x| \leq (m+2)/2$. Finally, if $x^2 = y$, then $|x|^2 = |y|$, which proves the equivalence. $\square$

This reduction also shows why the new h^* -real-rootedness theorem does not by itself settle the disk question: an additional root-location statement for the Ehrhart transform would still be needed. The argument below obtains the opposite conclusion from the snake-specific exact coefficients.

3. THE LENGTH-10 COUNTEREXAMPLE

Direct exact enumeration for $\mathbf{w}_{10}$ gives 13,860 linear extensions and

$$h^*(\mathbf{w}_{10}; z) = 1 + 21z + 181z^2 + 833z^3 + 2241z^4 + 3653z^5 \\ + 3653z^6 + 2241z^7 + 833z^8 + 181z^9 + 21z^{10} + z^{11}.$$

This is also an independent source-level check: Theorem 4.8 of [2] gives, for a regular word of length 10,

$$h^*(\mathbf{w}_{10}; z) = \sum_{i=0}^{11} D(11-i, i) z^i,$$

and evaluation of the Delannoy numbers gives exactly the displayed vector and sum 13,860. The reconstruction is checked independently from order-ideal multichains in Section 5.1. Substitution in (3), followed by exact integer factorization, gives

$$(4) \quad 24!L(\mathbf{w}_{10}; x-7) = 36 \prod_{k=0}^6 (x^2 - k^2) Q(x^2),$$

where

$$(5) \quad Q(y) = 385y^5 + 25789y^4 + 923223y^3 + 10769815y^2 + 70801492y + 23924096.$$

Proposition 3.1. *The polynomial Q has exactly one zero y_0 , counted with multiplicity, in*

$$\left| y_0 - \left(-\frac{53}{2} + 29i \right) \right| < \frac{1}{6}.$$

Every point of this disk has modulus greater than 36.

Proof. Put $c = -53/2 + 29i$ and write $Q(c+u) = \sum_{j=0}^5 a_j u^j$. Exact Gaussian-rational evaluation gives the certified bounds

$$\begin{array}{lll} |a_0| < 151000000, & |a_1| > 1060000000, & |a_2| < 103000000, \\ |a_3| < 3750000, & |a_4| < 61300, & |a_5| \leq 385. \end{array}$$

On $|u| = 1/6$, the lower bound for $|a_1 u|$ exceeds the sum of the upper bounds for all other terms by the strictly positive rational number

$$\frac{177200631815}{7776}.$$

Rouché's theorem therefore gives the same number of zeros as $a_1 u$ in the disk, namely one. Moreover,

$$|c|^2 - \left(36 + \frac{1}{6} \right)^2 = \frac{2117}{9} > 0.$$

The reverse triangle inequality shows that the whole isolating disk lies strictly outside $|y| = 36$. $\square$

Choose either x_0 with $x_0^2 = y_0$ and set $\rho_{10} = x_0 - 7$. Equation (4) gives $L(\mathbf{w}_{10}; \rho_{10}) = 0$, whereas

$$|\rho_{10} + 7| = |x_0| = \sqrt{|y_0|} > 6.$$

This proves Theorem 1.1(2). For orientation only, one choice is

$$y_0 \approx -26.4223522355 + 28.8791146449i, \quad \rho_{10} \approx -4.4780703509 + 5.7255987801i,$$

with $|\rho_{10} + 7| \approx 6.2564055612$. None of these decimals is used in the proof.

4. THE REPORTED LENGTH-9 BASELINE

An independent reconstruction for $\mathbf{w}_9$ enumerates 5,741 linear extensions and yields

$$h^*(\mathbf{w}_9; z) = 1 + 19z + 145z^2 + 575z^3 + 1289z^4 + 1683z^5 \\ + 1289z^6 + 575z^7 + 145z^8 + 19z^9 + z^{10}.$$

It also agrees term by term with the source's regular-snake formula $\sum_{i=0}^{10} D(10-i, i)z^i$. Let $u = 2t + 13$ and $s = u^2$. Exact expansion of (3) gives

$$(6) \quad 2^{22} 22! L\left(\mathbf{w}_9; \frac{u-13}{2}\right) = \prod_{k \in \{1, 3, 5, 7, 9, 11\}} (u^2 - k^2)(u^2 - 1)R(u^2),$$

where

$$(7) \quad R(s) = 5741s^4 + 1132460s^3 + 121839662s^2 + 3606687468s + 64103928525.$$

Set $c = -82 + 91i$. The exact Taylor coefficients of $R(c + v) = \sum_{j=0}^4 a_j v^j$ are

$$\begin{aligned} a_0 &= -3538359412 + 2090315136i, \\ a_1 &= 12454208952 - 3678772552i, \\ a_2 &= -210377920 - 204910524i, \\ a_3 &= -750588 + 2089724i, \quad a_4 = 5741. \end{aligned}$$

Their integer norm comparisons imply

$$\begin{aligned} |a_0| &< 4110000000, \quad |a_1| > 12980000000, \quad |a_2| < 294000000, \\ |a_3| &< 2221000, \quad |a_4| \leq 5741. \end{aligned}$$

On $|v| = 1/2$, the certified lower bound for the linear term exceeds the upper bound for all remaining terms by

$$\frac{36899552259}{16} > 0.$$

Thus Rouché's theorem isolates one zero s_0 in $|s_0 - (-82 + 91i)| < 1/2$. The entire disk is outside $|s| = 121$, since

$$|c|^2 - \left(121 + \frac{1}{2}\right)^2 = \frac{971}{4} > 0.$$

For $u_0^2 = s_0$ and $\rho_9 = (u_0 - 13)/2$, equation (6) gives $L(\mathbf{w}_9; \rho_9) = 0$ and

$$|\rho_9 + 13/2| = \frac{|u_0|}{2} = \frac{\sqrt{|s_0|}}{2} > \frac{11}{2}.$$

This proves Theorem 1.1(1). It does not identify the first failing length; it proves only that failure occurs no later than length 9.

5. INDEPENDENT EXACT VERIFICATION

The theorem is accompanied by two primary Rouché verifiers and a genuinely different length-10 audit. All use the Python standard library and exact integers or rational numbers. They reject unknown or missing fields and are pinned to the declared word, length, dimension, center, and radius.

The Rouché verifiers reconstruct the source cover relations, enumerate all linear extensions under a natural labeling (using the standard order-polytope descent formula [3]), rebuild $d!L(t)$ from (3), check the displayed factorizations and Taylor coefficients, and then check every strict rational inequality. They import neither discovery code nor discovery output. Structured tests

alter the word, endpoint, coefficient arrays, Taylor data, and Rouché bounds; both verifiers reject all eight corrupted inputs.

5.1. A different length-10 route. The independent audit starts from a six-field input containing only the word and target parameters. It enumerates the 46 order ideals of $P(\mathbf{w}_{10})$. Repeated subset convolution counts multichains and hence reconstructs $L(0), \dots, L(24)$; finite differences recover h^* without linear extensions. It then rebuilds (4).

For an independent root count, apply the Cayley map

$$y = 36 \frac{z+1}{z-1}, \quad \operatorname{Re} z = \frac{|y|^2 - 36^2}{|y - 36|^2}.$$

After removal of positive content, $(z-1)^5 Q(36(z+1)/(z-1))$ is

$$P(z) = 3943732660z^5 + 8365445947z^4 + 6584369218z^3 \\ + 2899674560z^2 + 1123599050z + 362656325.$$

The exact first column of its Routh table is

$$\begin{array}{r} 3943732660, \quad 8365445947, \quad \frac{43645643522626529846}{8365445947}, \\ \frac{388911394093047149050480765}{283413269627444999}, \quad -\frac{3013830289052014004207375413729280}{7071116256237220891826923}, \quad 362656325. \end{array}$$

There are two sign changes and no singular row, so P has exactly two roots in the open right half-plane. Since every coefficient of P is positive, neither root is positive real; they form a nonreal conjugate pair. The Cayley identity therefore gives a conjugate pair of zeros of Q with $|y| > 36$. This route shares no polynomial coefficients with its serialized input and does not use Rouché's theorem.

5.2. Reproduction commands and hashes. From the archive root, the decisive commands are

```
/opt/anaconda3/bin/python -I -B discovery/breaker_verify_counterexample.py \
  experiments/breaker_counterexample_certificate.json
/opt/anaconda3/bin/python -I -B audit/independent_routh_verifier.py \
  certificates/routh_audit_input.json
/opt/anaconda3/bin/python -I -B \
  discovery/breaker_length9_verify_counterexample.py \
  experiments/breaker_length9_counterexample_certificate.json
/opt/anaconda3/bin/python -B discovery/breaker_test_rejections.py
/opt/anaconda3/bin/python -B discovery/breaker_length9_test_rejections.py
/opt/anaconda3/bin/python -B tests/test_independent_routh_verifier.py
/opt/anaconda3/bin/python -B src/discover_snake_roots.py --max-length 10 \
  --output experiments/exact_scan_length0_10_anaconda.json
```

The certificate/verifier SHA-256 pairs are, respectively,

```
length 10 Rouché: d74f094c...ae68/7bcd9e2...bb73,
length 10 Routh: 5327157f...8fac/a7ec89fe...8275,
length 9 Rouché: a7a3f1b7...4cf5/f3abc48b...e38ed.
```

The full hashes and the files they bind are recorded in the accompanying manifest.

6. SCOPE, NOVELTY, AND DISCLOSURE

The proof needs a single length-10 word; it does not rely on symmetry quotienting or on a floating-point root scan. A standard-library exact scan visited every literal word of lengths 0 through 10. It reports no outside word through length 8, the two complementary regular words at length 9, and six literal words at length 10. The scan was repeated under two Python runtimes with identical mathematical fields. These records are supplementary context rather than a premise of Theorem 1.1; in particular, we do not promote the discovery scan to an independently audited minimality theorem. No reversal quotient was used, while complement pairs were checked directly.

The public-source search was frozen and repeated after the distinctive words and polynomial coefficients were known. Through August 29, 2026, it located the original paper, the Braun–Jal solution of the h^* half, and other work citing generalized snake posets, but no earlier proof, counterexample, or public certificate for the corrected disk statement. This is a bounded database-search report, not a claim of absolute priority.

OpenAI Codex assisted with literature retrieval, code generation, proof auditing, and manuscript preparation. The mathematical endpoint is bound to the exact reconstructing programs and serialized inputs described above. Floating-point arithmetic was used only to choose and display convenient isolating disks and decimal approximations; it is not used by the proof verifiers. No Lean, Coq, Isabelle, or other proof assistant was used.

DATA AND CODE AVAILABILITY

The source archive accompanying this manuscript contains the formal problem statement, literature ledger, exact certificates, full length-10 discovery records, three reconstructing verifiers, mutation tests, audit reports, and a SHA-256 manifest. The verifiers require only Python 3 and its standard library.

REFERENCES

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INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY, KUALA LUMPUR, 56000, MALAYSIA

Email address: zijianzeng@foxmail.com; 1002266693@ucsiuniversity.edu.my